

Prakash Mani Patel

Varanasi, Uttar Pradesh

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🌐 LinkedIn 🖥️ Hackerank 🐙 GitHub



Skills

Languages/Technologies: C, C++, Java, Python

Web Development: HTML, CSS, Streamlit, Git, GitHub, Vercel

Machine Learning/AI: NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch, OpenCV, Jupyter Notebook

Databases: MySQL, MongoDB

Relevant Coursework: Data Structures and Algorithms, Database Management Systems, Object-Oriented Programming, Operating Systems, Computer Networks

Technical Skills: Machine Learning, Deep Learning, Data Analysis, Web Development, Software Engineering

Education

Bachelor of Technology in Computer Science and Engineering, Jaypee University
of Engineering and Technology, Guna, Madhya Pradesh **Aug 2022 - May 2026**

- CGPA: 7.9

Class XII, St. Francis School, Varanasi, UP **Apr 2019 - Mar 2021**

- Percentage: 83.25%

Class X, St. Francis School, Varanasi, UP **Apr 2018 - Mar 2019**

- Percentage: 84.6%

Projects

FlavorMatch - Food Recommendation System **Jan 2025 – Apr 2025**

- Built a food recommendation system using content-based, collaborative, and hybrid filtering, achieving 85% top-N precision on a dataset of 7100+ recipes.
- Applied TF-IDF vectorization, cosine similarity, and user-item interaction modeling to capture recipe and user preference patterns.
- Implemented dimensionality reduction on 500+ flavor vectors, reducing model latency by 35% and boosting recommendation accuracy by 20%.
- Engineered a user profiling pipeline based on dietary, flavor, and time preferences, increasing recommendation relevance by 40%.
- Enhanced system efficiency through vectorized operations and sparse matrices, cutting computational overhead by 30%.
- Designed and deployed a Flask API backend on Vercel for real-time recommendations, supporting 100+ concurrent users.

Customer Churn Prediction - E-Commerce **Aug 2024 - Oct 2024**

- Developed a machine learning model to predict customer churn for an e-commerce platform, achieving 85% classification accuracy using Logistic Regression, Random Forest, and XGBoost.
- Engineered and selected key behavioral and transactional features, boosting model precision and recall metrics by 20%.
- Performed hyperparameter optimization with GridSearchCV and RandomizedSearchCV, improving model robustness and generalization.
- Visualized feature importance and model insights using Matplotlib to support data-driven decision-making.
- Conducted extensive model evaluation using confusion matrix, ROC-AUC analysis, and cross-validation, ensuring high reliability across diverse customer segments.
- Delivered actionable insights, reducing churn by 15% and increasing customer lifetime value by 10%.

Certifications

- The Complete Full-Stack Web Development Bootcamp (Udemy)
- Machine Learning Foundation (Infosys Springboard)
- Data Structures and Algorithms in Python (Infosys Springboard)